Kate Leemann
ITAI 1378
Professor McManus
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A01 ITAI 1378 Colab and other Tools Practice Run

Link to public github repository: https://github.com/SCheeses/jupyter-exploration

What I Did

In this lab, I delved deeper into using GitHub and Jupyter Notebooks as tools for programming and data science work. I began by logging into my GitHub account and creating a new repository named jupyter-exploration, initializing it with a README file. I edited the README file and made my first commit to the main branch.

Next, I set up my working environment for Jupyter Notebooks. Rather than using an online notebook service as I have in previous semesters, I wanted to challenge myself by working with Visual Studio Code. I also installed the Jupyter extension so I could become more familiar with tools that are commonly used in industry. I downloaded all of the required extensions and created my first notebook file, My_First_Notebook.ipynb. Within that notebook, I created a Markdown cell containing descriptive text and a code cell containing a simple Python command that printed "Hello, World!" to the screen. I executed both cells and observed the output.

After completing the notebook, I saved the file and uploaded it to my GitHub repository. I committed the notebook file with an appropriate commit message and pushed the changes to GitHub.

What I Learned

This lab introduced me to several important concepts related to software development and data science workflows. I have previously worked with Google Colab and the GitHub online interface for my repositories and school assignments. Working with VS Code was not necessarily more complicated, but the steps were not as apparent. I initially ran into an issue where Python was not working in my notebook.

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After troubleshooting, I realized that I had installed Python-related modules but not the Jupyter Notebook extension. As a result, I could create an .ipynb file, but I could not execute code within it until the correct extension was installed.

One challenge I encountered involved configuring Git and understanding how GitHub, GitHub Desktop, and Visual Studio Code work together. Initially, it was not obvious which tasks should be performed in the browser versus on my local computer. When I checked my online repository in the browser, I found a file that I had not intentionally added. This file was titled ".DS_Store." I had to do some research to figure out what it was and learned that it is a file created by macOS that is normally hidden from the user. I also had to learn why it appeared in my repository and how to prevent it from being tracked. I deleted the file directly through GitHub. By researching the issue and following the recommended Git workflow, I was able to understand the purpose of .gitignore files and will use them in the future to improve the organization of my repositories and avoid committing unintended files.

Overall, this lab provided a great introduction to tools used in software development, data science, and machine learning projects.

Questions or Comments

This lab was a useful introduction to GitHub and Jupyter Notebooks. One area I would like to explore further is how professional developers use Git and GitHub in larger team environments. For example, I am somewhat familiar with README files, but I would like to learn more about what information should be included in them and how they are typically structured. I want to implement good practices, even in assignments like this one, so that my workflow more closely reflects how projects are managed in a professional development environment.